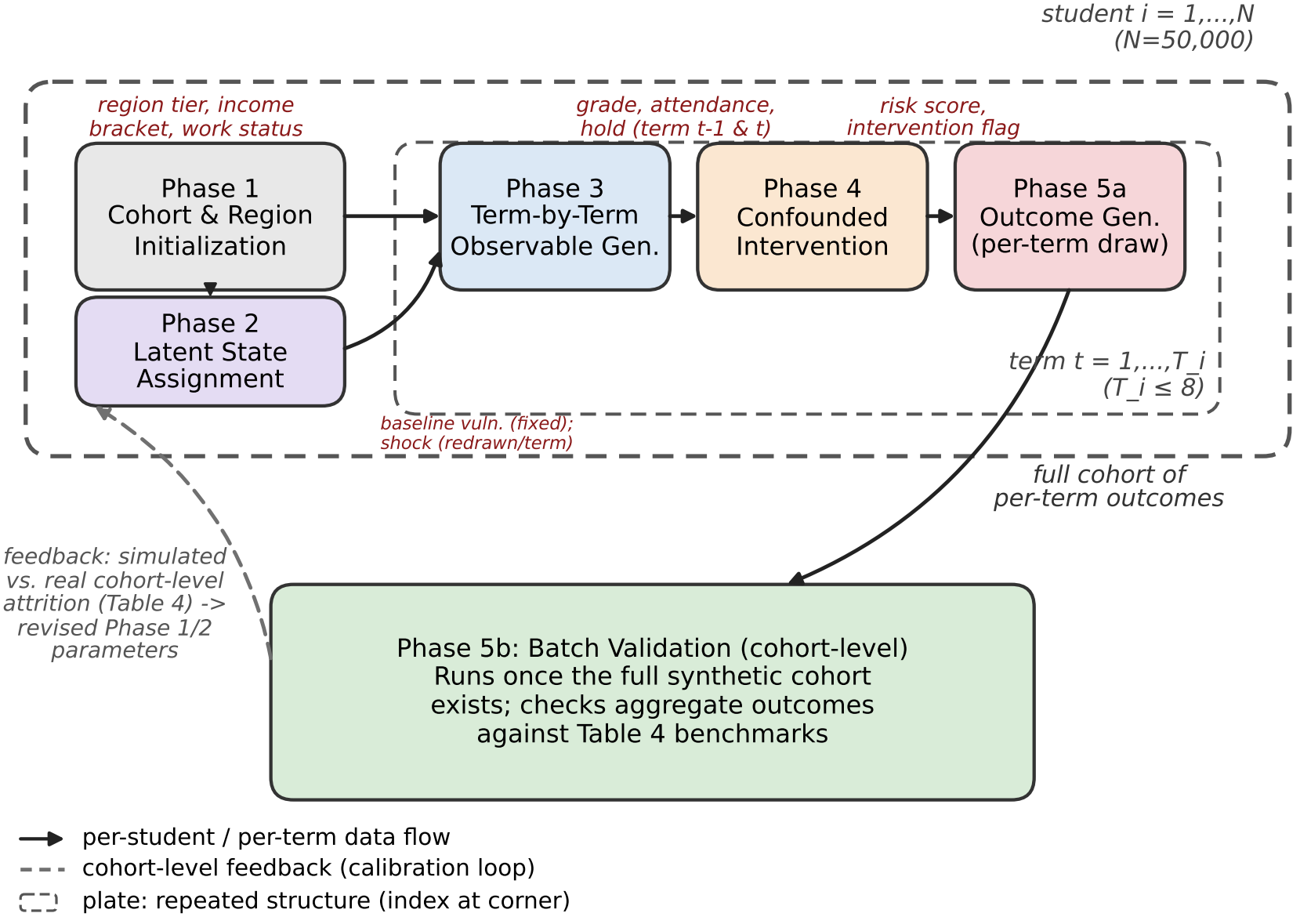
student i = 1,...,N
(N=50,000)
region tier, income bracket, work status
grade, attendance, hold (term t-1 & t)
risk score, intervention flag
Phase 1
Cohort & Region
Initialization
Phase 2
Latent State
Assignment
Phase 3
Term-by-Term
Observable Gen.
Phase 4
Confounded
Intervention
Phase 5a
Outcome Gen.
(per-term draw)
term t = 1,...,T_i
(T_i ≤ 8)
baseline vuln. (fixed);
shock (redrawn/term)
full cohort of
per-term outcomes
feedback: simulated
vs. real cohort-level
attrition (Table 4) ->
revised Phase 1/2
parameters
Phase 5b: Batch Validation (cohort-level)
Runs once the full synthetic cohort
exists; checks aggregate outcomes
against Table 4 benchmarks
per-student / per-term data flow
cohort-level feedback (calibration loop)
plate: repeated structure (index at corner)